COMPREHENSIVE DEVELOPMENT PLAN

Novelty Lens

From Patent Search Tool to Auditable Novelty Search Platform

A product and engineering roadmap for semantic and hybrid retrieval, human review, reference management, evidence comparison, and professional novelty reporting.

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Based on the current repository architecture and analysis of the Eureka by Patsnap workflow

Executive Summary

Recommendation

Deliver Novelty Lens through four connected product releases: invention review, hybrid retrieval, reference and evidence management, and an auditable novelty report. The primary investment should be retrieval quality and evidence traceability rather than visual imitation of Patsnap.

The project already has a strong technical foundation: FastAPI, Angular 22, a single LangGraph pipeline, real SSE progress, LLM-based invention understanding, live patent search, relevance validation, embedding-based ranking, explanations, PostgreSQL history, and a written Qdrant semantic branch. The main gap is that the current experience moves directly from input to final report instead of exposing a reviewable, iterative search process.

The target platform should let users review what the system understood, approve atomic technical features, inspect keywords and classifications, follow each search strategy, curate references, and receive a feature-by-feature evidence matrix with verifiable quotations and source locations.

Target Outcomes

- Increase recall of meaningful prior art without hiding precision loss or using misleading expansions.
- Separate retrieval from legal assessment: similarity is not novelty, and no result does not prove novelty.
- Introduce human review before expensive or consequential search stages.
- Provide complete provenance for every query, reference, feature judgment, quotation, and report conclusion.
- Preserve the zero-tolerance policy: required-step failure must produce a clear error, never fabricated or silently degraded output.

Scope

This plan covers product workflow, data, retrieval architecture, semantic search, lexical search, user interfaces, persistence, testing, operations, security, metrics, and delivery sequencing. USPTO integration remains out of scope because the project has no approved API access. No unavailable data provider is assumed.

1. Current-State Baseline

Area	Current Capability	Gap to Target
Orchestration	LangGraph is the only pipeline; progress is	Persisted review checkpoints and resumable

Area	Current Capability	Gap to Target
	streamed through SSE.	execution.
Invention understanding	LLM understanding and query generation.	Atomic features, core-feature selection, relationships, and editable IPC/CPC.
Patent search	Google Patents through SerpAPI; EPO implementation is disabled.	Scalable hybrid retrieval, families, citations, and a sustainable provider.
Semantic retrieval	Qdrant, indexing, semantic search, and fusion are implemented but disabled.	Calibrated Qwen retrieval, multi-vector indexing, and reranking.
Validation and ranking	LLM relevance validation and embedding ranking.	Feature-level evidence with exact quotations and locations.
Persistence	Completed runs and final matches are stored in PostgreSQL.	Drafts, reviews, strategies, query runs, references, and evidence.
Frontend	Submit, running, report, history, login, and error screens.	A multi-stage search workspace with review and approval actions.
Non-patent literature	Not part of the current pipeline.	Verified papers and web sources with strict anti-fabrication controls.

2. Governing Design Principles

Evidence first: No feature match is displayed without a source quotation, a stable locator, and a link to the source document.

Human review at consequential decisions: Users approve the technical summary, features, and search plan before broad external retrieval or deep analysis.

Retrieval is not judgment: Semantic and Boolean matches are candidates; novelty assessment requires document analysis and date constraints.

No silent fallback: Required providers and steps fail clearly. Optional sources must be declared optional before the run begins.

Reproducibility: Each run records queries, strategy versions, model versions, providers, dates, and configuration.

Family-aware results: Patent families are consolidated so result volume does not masquerade as evidence diversity.

Evaluation before expansion: No retrieval strategy enters production without benchmark evidence for quality, latency, and cost.

3. Target User Journey

1. Enter the invention description, optional claims and attachments, cutoff date, and target jurisdictions.
2. Generate an editable technical solution summary with provenance back to the submitted text and attachments.
3. Extract atomic technical features; allow the user to split, merge, delete, edit, and flag core features.
4. Present search elements: keywords, related terms, entities, exclusions, and IPC/CPC classifications with relationship labels.
5. Propose a search plan with estimated queries, cost, and result limits; require approval before execution.
6. Run semantic, lexical, classification, family, citation, and optional non-patent searches in measured stages.
7. Create a unified reference list with family consolidation, add/remove/pin actions, and manual reference entry.
8. Build an evidence matrix linking each reference to each technical feature with a quotation and locator.
9. Generate a novelty report that separates single-reference disclosure, combined coverage, gaps, and limitations.

3.1 Human-in-the-Loop Editing and Approval Contract

- Edit or selectively regenerate the technical summary while preserving the original submission and every approved revision.
- Add, edit, delete, reorder, split, merge, and mark core technical features before retrieval begins.
- Add or exclude keywords, relationships, IPC/CPC classes, jurisdictions, and manually supplied references.
- Show the expected impact of each edit: which downstream search elements, strategies, and evidence cells become stale and require recomputation.
- Provide undo, revision history, and an audit trail recording who changed what and when; never overwrite an approved version silently.
- Persist waiting_for_review checkpoints so users can leave and resume without rerunning completed external or expensive stages.
- Require explicit approval at three MVP gates: invention summary and features, search plan and limits, and final reference selection before report generation.

Current status: these capabilities are planned for R1 and are not yet available in the present submit-running-report interface.

4. Target Architecture

End-to-End Data Flow

Input -> Normalize -> Technical Summary -> Feature Graph -> Search Elements -> Search Plan -> Hybrid Retrieval -> Family Consolidation -> Reranking -> Evidence Extraction -> Reference Review -> Novelty Assessment -> Report and Persistence

4.1 Proposed LangGraph Nodes

Node	Responsibility	Persisted Output
normalize_input	Normalize text, attachments, jurisdictions, and cutoff date.	normalized_input, attachments_manifest
summarize_solution	Create a technical summary without adding unsupported content.	technical_summary, provenance_spans
extract_features	Extract atomic features, relationships, and core candidates.	features, feature_graph
review_features	Human checkpoint with revision and approval.	approved_features, revision
build_search_elements	Generate terms, exclusions, entities, and IPC/CPC with reasons.	search_elements, exclusions
plan_search	Build strategies, ordering, budgets, and stop rules.	strategies, estimated_cost
semantic_retrieve	Retrieve candidates from a calibrated multi-vector index.	semantic_hits
lexical_retrieve	Run keyword, Boolean, and classification queries with counts.	query_runs, lexical_hits
track_family_citations	Retrieve families, citations, and representative documents.	family_hits, citation_hits
retrieve_npl	Retrieve verified papers and web resources when enabled.	papers, web_sources
fuse_dedup	Fuse ranked lists and remove publication/family duplicates.	candidate_set
rerank	Apply accurate reranking to the limited candidate set.	ranked_candidates
extract_evidence	Map features to exact source quotations and locations.	evidence_cells
review_references	Human selection, exclusion, and manual additions.	approved_references
assess_novelty	Assess single-reference disclosure, gaps, and limitations.	assessment, limitations
persist_and_report	Atomically store the completed run before emitting one result.	report_id, final_report

4.2 SSE Contract

Keep the existing event types progress, result, and error. Extend progress with stage, strategy, and optional counters. Review checkpoints should persist a waiting_for_review status and

resume through explicit endpoints; they must not emit a partial report. The stream must still finish with exactly one result event after persistence succeeds.

- Update backend phases, frontend PHASE_ORDER, persistence, and contract tests together.
- Any new reference field must be produced, persisted, declared in Angular, and covered by a history round-trip test.
- Preserve verbatim provider errors without exposing credentials or query-string secrets.

5. Semantic Retrieval Design

5.1 Corpus and Indexing

Retrieval quality depends more on corpus coverage and document representations than on the vector database brand. For the MVP, build a versioned cache from externally retrieved, licensed, and verifiable patent documents; a global local corpus is not required. Never overwrite a full-text representation with an abstract-only record.

Representation	Indexed Content	Purpose
document_overview	Title, abstract, and a source-grounded technical summary.	Broad topical retrieval.
independent_claim	Each independent claim as a separate unit.	Retrieve close legal and technical structures.
claim_chunk	Dependent claims grouped with their parent claim.	Capture detailed limitations.
description_chunk	Overlapping passages with paragraph locators.	Enable precise evidence extraction.
feature_summary	Validated technical features derived from the document.	Match complex technical concepts.

5.2 Query and Reranking Flow

1. Embed the complete technical summary for broad conceptual retrieval.
2. Embed each core feature separately to retrieve rare mechanisms hidden by the overall topic.
3. Apply metadata filters for cutoff date, jurisdiction, document type, language, and full-text availability.
4. Fuse the separate ranked lists while retaining the strategy that contributed each candidate.
5. In free MVP mode, retrieve 50-100 lightweight candidates in total, consolidate families, semantically rerank the best 10-15 locally, and send only the best 3-5 references to deep evidence extraction.

5.3 Qwen and Qdrant Calibration Gate

- Build a benchmark of inventions and known examiner or expert references with valid cutoff dates.

- Measure score distributions for related and unrelated text; do not reuse the stale 0.6 threshold.
- Test chunk size, overlap, and maximum chunks against recall@30, latency, and cost.
- Validate the Qwen3 query instruction empirically before enabling semantic search.
- Recreate the Qdrant collection whenever vector dimensions or document embedding strategy changes.

5.4 Free MVP Retrieval Funnel

- Candidate generation: retrieve 50-100 records in total using lightweight metadata such as title, abstract, dates, classifications, and source link.
- Cheap filtering: enforce the cutoff date, remove publication duplicates, consolidate patent families, and reject clearly unrelated records before full-text retrieval.
- Local semantic stage: use the existing local embedding capability to rerank only the best 10-15 candidates; do not require a global patent vector corpus for the MVP.
- Deep evidence stage: fetch claims and relevant description passages only for the best 3-5 references and extract quotations with stable locators.
- Adaptive stopping: stop when additional queries contribute no new useful families; allow the user to request an expanded search explicitly.
- Product disclosure: label this accurately as semantic reranking of externally retrieved candidates, not exhaustive semantic search across the global patent corpus.

6. Lexical, Classification, and Iterative Search

Do not hard-code four Boolean rounds merely to imitate Patsnap. Use a search planner that selects strategies based on features and prior results, then stops when marginal value falls or the approved budget is reached.

Strategy	Typical Formula	Use	Stop Signal
B1 Topic	Title/Abstract/Claims plus IPC	Broad subject recall.	Manageable results and valid references.
B2 Features	Related-term blocks for each feature	Find specific mechanisms.	Few new useful families after deduplication.
B3 Intersections	S1 AND S2 AND S3	Narrow technical combinations.	Useful volume without false zero-result certainty.
B4 Result learning	Terms and classes from strong seed patents	Discover industry vocabulary.	Marginal reference gain approaches zero.
Tracking	Family plus cited and cited-by	Expand around selected seeds.	Depth one by default; deeper only with approval.

Expansion Quality Guard

Every related term must carry a relationship type, source, and confidence. USB and Ethernet must not be treated as synonyms; radio frequency must not silently expand into every wireless technology. Broader and narrower relationships may be used only when explicitly labeled and justified.

7. Data Sources and Sustainability

The current SerpAPI allowance is suitable for limited development but not for a professional multi-strategy platform. A complete run consumes roughly 25 requests from a monthly allowance of 250. Select and contract a provider before scaling retrieval breadth.

Requirement	Minimum Capability	Acceptance Test
Search fields	Title, abstract, claims, description, IPC, and CPC.	Run fixed queries and compare stable result counts.
Full text	Claims and description with stable paragraph locations.	Evidence quotation matches the original source.
Families	Simple or extended family and representative member.	Consolidate a cross-jurisdiction sample.
Citations	Cited and cited-by with explicit direction.	Verify a manual sample and source links.
Dates	Priority, filing, and publication dates.	Cutoff filtering returns no later documents.
Quota and cost	Capacity for benchmark and production use.	Model monthly workload and rate-limit behavior.
License	Internal storage and indexing rights.	Written legal approval before ingestion.

8. Proposed Data Model

PostgreSQL remains the source of truth for runs, reviews, and reports. Qdrant is a rebuildable retrieval index and must not become the authoritative evidence store.

Entity	Key Fields	Notes
search_runs	status, cutoff_date, graph_version, model_versions	Support draft, waiting, running, completed, and failed states.
technical_summaries	text, revision, source_spans, approved_at	Immutable revisions; no silent replacement.

Entity	Key Fields	Notes
features	feature_id, text, core, parent, relation	Stable ordering and revision history.
search_elements	term, relationship, source, confidence	Synonym, broader, narrower, class, or exclusion.
search_strategies	type, query, fields, status, result_count	Complete audit trail for reproducibility.
references	publication, family, dates, source, selected	Unified patent and NPL model.
evidence_cells	feature, reference, judgment, quote, locator	No judgment without quotation and locator.
reports	assessment, limitations, generated_at	Immutable completed report.

9. Required Product Surfaces

Invention Review

Edit the technical summary, inspect provenance, approve, or regenerate a defined section.

Feature Review

Edit, split, merge, remove, reorder, and flag core features; warn about generic statements.

Search Elements

Review related terms, exclusions, entities, and classifications with source and relationship.

Search Monitor

Display strategy tree, exact query, result count, duration, cost, and error state.

Reference Workspace

Patent, paper, and web tabs; family groups; pin, remove, add, and manually enter references.

Evidence Matrix

Feature rows and reference columns with disclosed, partial, not disclosed, and evidence drill-down.

Novelty Report

Assessment, limitations, recommendations, source links, and auditable export.

10. Delivery Roadmap

Release	Objective	Primary Deliverables	Exit Gate
R0 Baseline	Establish measurable quality before change.	Benchmark, telemetry, phase alignment, failure matrix.	Reproducible baseline report.
R1 Review Workspace	Make invention understanding reviewable.	Editable summary, feature and search-element controls, revision history, approvals, persistence, and resume.	Users add, edit, delete, undo, approve, leave, and resume without losing state.
R2 Hybrid Retrieval	Improve recall with transparent strategies.	Calibrated Qdrant, multi-vector index, planner, fusion, reranking.	Recall and precision targets met on benchmark.
R3 Evidence Workspace	Convert results into auditable evidence.	Reference list, families, citations, evidence matrix, quotations.	Every judgment has a source and locator.
R4 NPL and Report	Deliver complete review-ready reports.	Papers, web sources, novelty logic, limitations, exports.	Expert review passes benchmark cases.
R5 Scale and Governance	Operate sustainably in an enterprise setting.	RBAC, quotas, observability, retention, audit logs.	Security and SLO approval.

11. First 90 Days

Period	Work	Output
Weeks 1-2	Build benchmark, measure current pipeline, assess providers, complete failure matrix.	Baseline and data decision.
Weeks 3-5	Add versioned summary, feature, and search-element models; migrations; impact analysis; review and resume endpoints.	R1 backend.
Weeks 6-7	Build review screens with add/edit/delete/split/merge/undo controls, connect SSE and history, and add Angular checkpoint tests.	Completed R1 frontend.
Weeks 8-10	Calibrate Qwen/Qdrant, enable semantic indexing and fusion behind evaluation controls.	Hybrid retrieval beta.
Weeks 11-12	Implement search planner, query audit, result counts, and family deduplication.	Search monitor beta.
Week 13	Review metrics and defects; decide whether to begin the Evidence Workspace.	R3 go/no-go decision.

12. Prioritized Backlog

Priority	Item	Value
P0	Align frontend PHASE_ORDER with backend phases, including filter.	Accurate progress experience.

Priority	Item	Value
P0	Create a cutoff-aware benchmark with known X/Y references.	Objective before/after measurement.
P0	Select a patent data provider and confirm indexing rights.	Remove the largest delivery risk.
P0	Persist summaries, features, strategies, and resumable state.	Foundation for human review.
P0	Calibrate Qwen threshold and query instruction.	Safe semantic-search activation.
P1	Index claims and description chunks as multiple vectors.	Improved detailed-feature recall.
P1	Add search planning and query-result telemetry.	Transparent iterative retrieval.
P1	Add family consolidation and citation tracking.	Less duplication and stronger seed expansion.
P1	Extract feature evidence with stable quotation locators.	Auditability.
P2	Add verified NPL providers and a unified reference model.	Non-patent prior art coverage.
P2	Add user ownership and scope destructive history operations.	Multi-user readiness.
P2	Add explicit freshness-aware caching.	Lower cost without silent reuse.
P0	Implement free-mode adaptive funnel: 50-100 lightweight candidates, 10-15 local semantic analyses, and 3-5 deep evidence reviews.	Control latency and cost while preserving an upgrade path.

13. Test and Evaluation Strategy

13.1 Retrieval Metrics

- Recall@10, Recall@30, and Recall@100 for known references; MRR for the first X or Y document.
- Precision@30 after reranking; post-cutoff leakage must be zero.
- Family diversity and useful new references contributed by each strategy.
- Ablation analysis for semantic, Boolean, classification, and citation retrieval.

13.2 Evidence Metrics

- Feature-disclosure precision and recall against expert annotations.
- Quotation grounding rate: the cited text must actually support the judgment.
- Locator validity: the system must open the correct claim or paragraph.
- Fabricated reference rate must remain zero; unverifiable references fail the stage.

13.3 Failure Matrix

Test LLM failure, truncation, malformed JSON, embedding outage, Qdrant outage, provider outage, 429 rate limits, 402 quota exhaustion, PostgreSQL failure, missing full text, zero results, and user cancellation during every stage. The only acceptable outputs are a clear error or a genuine zero-result response, never a partial report.

14. Security and Governance

- Replace the public demo account and default AUTH_SECRET before any deployment beyond localhost.
- Add user_id or tenant_id to runs and references; scope history deletion to the authenticated owner.
- Never expose provider keys in URLs, logs, or SSE errors; retain secret-leak regression tests.
- Define retention and deletion policies for invention text, claims, and attachments.
- Record approvals and edits in an audit trail without storing model chain-of-thought.
- Review corpus licensing before storing full text or derived embeddings.

15. Performance and Cost

The goal is not maximum result volume or minimum latency at any cost; it is the strongest prior-art set within a transparent budget. Free MVP mode uses an adaptive funnel: 50-100 lightweight candidates, 10-15 local semantic analyses, and 3-5 deep evidence reviews. Estimate cost before execution, cap each strategy, deduplicate before full-text calls, and parallelize independent work only when provider quotas and failure semantics remain correct.

Area	Optimization	Protection
Embeddings	Use measured small concurrent batches.	Timeout and vector-dimension checks.
Full text	Fetch only candidates that pass prefiltering.	Do not analyze abstract-only records as full evidence.
LLM	Bounded batches and structured outputs.	Treat truncation as failure; do not fabricate repairs.
Search APIs	Query budgets and early deduplication.	Check remaining quota before benchmarks.
Caching	Reuse only dated, source-backed artifacts.	Expose freshness; never silently reuse a report.

16. Key Risks and Mitigations

Risk	Impact	Mitigation
Small or biased corpus	Important prior art is missed.	Broad provider coverage, benchmark, and coverage dashboard.
Overbroad term expansion	Noise and unnecessary cost.	Relationship taxonomy, counts, approval, and exclusions.
Misleading similarity	Topical rather than mechanical matches.	Reranking, feature evidence, and independent-claim weighting.
Unsustainable quota	Testing and production stop.	Budgets, mock mode, quota monitoring, and suitable provider contract.
Overstated legal conclusion	False confidence in novelty.	Explicit limitations and separation of retrieval from legal judgment.
SSE contract drift	Broken progress, reports, or history.	Contract tests and coordinated producer/persistence/frontend changes.
Invention data leakage	Confidentiality and IP exposure.	Access control, retention, encryption, and vendor review.

17. Team and Responsibilities

- Product and IP lead: define examiner-style workflow, acceptance criteria, and legal boundaries.
- Search and ML engineer: embeddings, indexing, retrieval, reranking, and evaluation.
- Backend engineer: LangGraph, APIs, persistence, providers, and SSE.
- Frontend engineer: review workspace, evidence matrix, accessibility, and exports.
- Patent reviewer: annotate benchmark cases and validate evidence quality.
- QA and security support: failure injection, secrets, tenancy, and performance testing.

18. Release Readiness Criteria

- The agreed benchmark shows documented improvement over baseline.
- Every result records its source, strategy, family, and required dates.
- Every feature judgment has a valid quotation and locator or remains unresolved.
- Failure and absence tests produce no invented references, papers, recommendations, or evidence.
- Runs resume after review without repeating completed external or deep-analysis stages.
- SSE contracts, persistence round trips, backend tests, and Angular build all pass.
- Light, dark, loading, error, empty, and review states receive visual QA.
- Cost limits, SLOs, retention, ownership, and access controls are documented and approved.

19. Decisions Required Before R2

1. Which patent-data provider will be primary, and what storage and indexing rights are granted?
2. What is the default cutoff date, and how will non-patent prior art dates be validated?
3. What benchmark size, expert labeling process, and success threshold will be used?
4. Are papers and web sources required or optional and explicitly labeled?
5. What cost and latency budget is acceptable for one novelty search?
6. Is the product an internal decision-support tool or a user-facing legal assessment product?

20. Conclusion

Recommended Sequence

Start with R0 and R1: measurement and invention review. Select the data provider in parallel. Then deliver R2 against a closed benchmark before building the Evidence Workspace. This avoids placing a large interface on top of an uncalibrated retrieval engine and preserves the project's strongest property: honest, traceable output.

Functional parity with Patsnap is feasible. Sustainable differentiation, however, will not come from the number of strategies or the sidebar design. It will come from measurable hybrid retrieval, disciplined query expansion, exact evidence, and a report that states both what the system found and what it could not establish.

Appendix A. Sources and Planning Inputs

- Repository architecture and constraints documented in CLAUDE.md as of the plan date.
- User-provided screenshots of the Eureka workflow: technical summary, feature extraction, search elements, semantic and Boolean search, citation tracking, and reference list.
- Patsnap Help Center, Eureka by Patsnap IP Walkthrough: structured multi-strategy retrieval and RAG/RAT description.
- Patsnap Eureka Novelty Search: keyword, concept, classification, jurisdiction expansion, and the published claim of more than 40 strategies.

Note: Patsnap does not publicly disclose its embedding model, vector dimensions, or exact reranking architecture. Any architectural interpretation of those details is an engineering inference, not a published fact.